Name: Ramani Jyothi Prasad

Student ID: A00096829

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PART 1: APPLIED PROBLEM-SOLVING ACTIVITY

MATHEMATICAL AND STATISTICAL ANALYSIS OF GLOBAL MDR/RR TUBERCULOSIS BURDEN USING PYTHON

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Task 1: Dataset Description

1.1 Data Origin and Source Explanation

Data Source: Tuberculosis data

The dataset used for analysis compiles country-level related to MDR/RR-TB from nations worldwide. It gives information on drug-resistant tuberculosis at the national level, including estimated resistance rates and associated epidemiological measures. The dataset was obtained in CSV format from a global health open-data platform and analysed using Python tools such as Pandas and NumPy. It holds well-structured numerical variables that make it possible to compare countries, conduct statistical analysis and probability modelling, and explore global patterns and trends in TB drug resistance. In addition, the organized structure of the data supports efficient preprocessing, visualization, and evaluation of statistical distributions within a typical data science workflow.

1.2 Description of Variables in the Dataset

The dataset contains variables associated with drug-resistant tuberculosis, where each row represents a country and the columns include indicators such as the estimated proportion of MDR-TB among new cases and previously treated cases. The data is organized well-structured numerical format, it supports statistical analysis, enables comparisons between countries, and helps to find patterns and trends in TB drug resistance across different nations.

1.3 Importance and Relevance of the Dataset

The given dataset is important because tuberculosis moves to be a major global health issue, mainly the growing challenge of multidrug-resistant tuberculosis (MDR-TB). Its importance lies in several aspects:

- It helps in analysing global trends and patterns of drug resistance.
- It supports the development of evidence-based public health policies and strategies.
- It allows the application of statistical methods and mathematical modelling for deeper analysis

Task 2: Mathematical Application

2.1 Linear Algebra – Vectors

```python
# STEP-1 LINEAR ALGEBRA - VECTOR OPERATIONS
# ========================================

# Load cleaned dataset
df = pd.read_csv("clean_tb_dataset.csv")

# Select variables from cleaned dataset
v_new = df["e_rr_pct_new"].values
v_ret = df["e_rr_pct_ret"].values

# Convert to numpy vectors
v_new = np.array(v_new)
v_ret = np.array(v_ret)

# Dot Product
dot_product = np.dot(v_new, v_ret)

# Vector Norms
norm_v_new = np.linalg.norm(v_new)
norm_v_ret = np.linalg.norm(v_ret)

# Cosine Similarity
cosine_similarity = dot_product / (norm_v_new * norm_v_ret)

# Create results table
metrics = ["Dot Product", "Norm of v_new", "Norm of v_ret", "Cosine Similarity"]
values = [dot_product, norm_v_new, norm_v_ret, cosine_similarity]

results = pd.DataFrame({"Metric": metrics, "Value": values})

print("VECTOR ANALYSIS SUMMARY")
display(results)
```

```
VECTOR ANALYSIS SUMMARY
```

	Metric	Value
0	Dot Product	218439.000000
1	Norm of v_new	316.069613
2	Norm of v_ret	873.080180
3	Cosine Similarity	0.791577

Figure 2.1.1: Vector Analysis of Treatment Outcomes Using Dot Product, Vector Norms, and Cosine Similarity

In this section, vectors from linear algebra are used to organize numerical data in a structured format. In this process, selected variables from the tuberculosis dataset are expressed as vectors, and each element represents an observation at the country level. Vector operations such as the dot product, vector norm, and cosine

similarity are applied to examine relationships and measure the magnitude of the data, producing numerical results. The outcomes of these operations are illustrated in Figure 2.1.1.

2.2 Linear Algebra – Matrices

```
# STEP-2 LINEAR ALGEBRA - MATRICES
# ================================

# Data matrix
X = df[["e_rr_pct_new", "e_rr_pct_ret", "e_inc_rr_num"]].values
print("Data Matrix: ")
print(X[:5])

# XᵀX matrix
XtX = X.T @ X
print("\nXᵀX:\n", XtX)

# Rank and Determinant
print("\nRank of X:", np.linalg.matrix_rank(X))
print("Determinant of XᵀX:", np.linalg.det(XtX))
```

```
Data Matrix:
[[   8   28 6400]
 [   7   24 5500]
 [   6   21 4800]
 [   5   18 4200]
 [   4   16 3600]]

XᵀX:
 [[       99900       218439     37970536]
 [      218439       762269    102107358]
 [    37970536    102107358 224245330559]]

Rank of X: 3
Determinant of XᵀX: 5.929681334964054e+21
```

Figure 2.2.1: Construction of Data Matrix and Computation of *X^ T,* Rank analysis, and Determinant

In data science, matrices are used to arrange multiple variables into an systemstic numerical structure that allows efficient mathematical analysis. In this evaluation, a data matrix **X** was created using selected tuberculosis indicators, where each row represents a country and each column represents a specific variable. This matrix format makes it possible to analyse several variables simultaneously.

As illustrated in **Figure 2.2.1**, the matrix operation X^T X was calculated to analyse the relationships between the variables. Also find rank analysis of the matrix was also evaluated to determine how many variables are linearly independent and to assess the amount of unique information present in the dataset. Finally, the

determinant of X^T X was computed to examine properties of the matrix, including linear dependence and numerical stability. These matrix operations provide deeper mathematical insight into the structure of the data and the relationships among the variables.

2.3 Descriptive Statistics and EDA

```python
# STEP-3 DESCRIPTIVE STATISTICS & EDA
# ===================================

# Select two key variables from dataset
var_new = df["e_rr_pct_new"]
var_ret = df["e_rr_pct_ret"]

# Compute Summary Statistics
summary_stats = df[["e_rr_pct_new","e_rr_pct_ret"]].agg(["mean","var","skew","kurt"])
summary_stats.columns = ["New Cases","Retreatment Cases"]
summary_stats.index = ["Mean","Variance","Skewness","Kurtosis"]

# Print Summary Results
print("SUMMARY STATISTICS")
display(summary_stats)
```

```
SUMMARY STATISTICS
```

	New Cases	Retreatment Cases
Mean	3.635655	15.186072
Variance	42.460267	194.153764
Skewness	3.392462	2.114921
Kurtosis	11.833467	5.350430

Figure 2.3.1: Summary Statistics (Mean, Variance, Skewness, Kurtosis) for RR-TB Percentages in New and Retreatment Cases

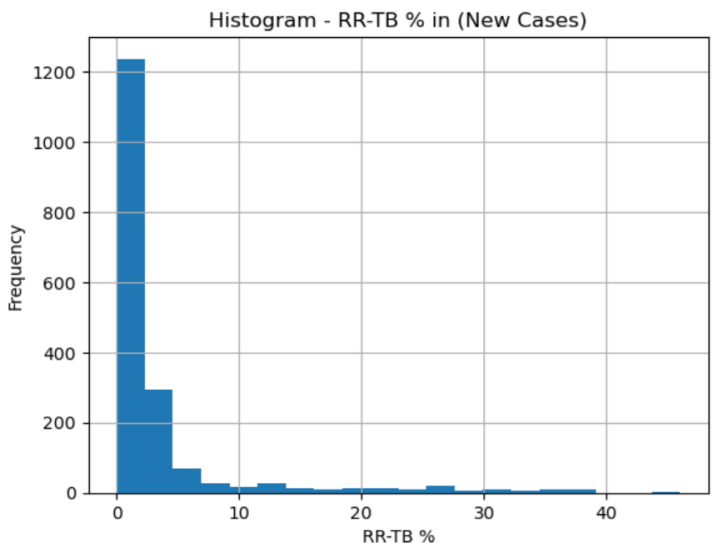
Figure 2.3.1 displays the descriptive statistics for the percentage of rifampicin-resistant tuberculosis (RR-TB) among both new and retreatment cases. These statistics measures such as the mean, variance, skewness, and kurtosis, which help summarise the main analysis of the dataset.

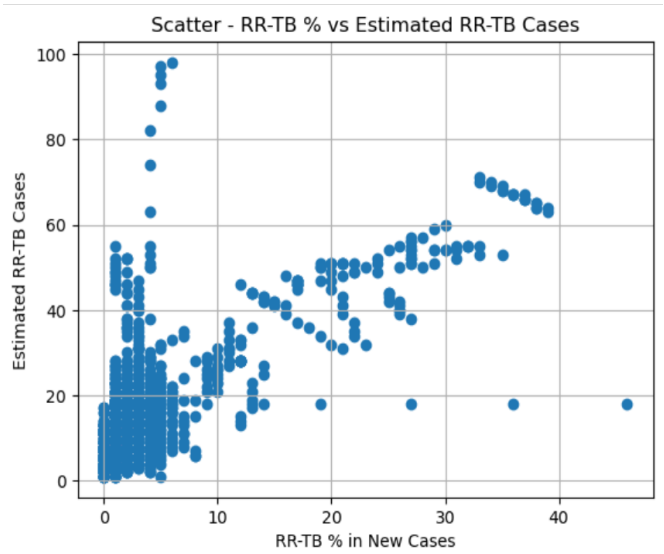
The mean represents the average percentage of RR-TB observed in each category, offering a general signal of the overall level of drug resistance. Variance measures how much the values differ from the mean, showing the level of variation in RR-TB percentages across the observations. Skewness is used to assess the symmetry of the distribution; positive or negative values depicts the data are skewed toward higher or lower values.

Kurtosis describes the shape of the distribution by indicating its peaked ness and the likelihood of extreme values.

These descriptive statistics provide an initial summary of the dataset and help highlight key distributional patterns, with the corresponding plots presented in above figure 2.3.1. This preliminary analysis is valuable for understanding the characteristics of the variables before applying more advanced statistical techniques, such as distribution checks and goodness-of-fit tests.

```
# Visualisation Graphs(Plots)
# --------------------------

# Plot configurations (title, xlabel, ylabel)
plots = [
    (plt.hist, v_new, "Histogram - RR-TB % in (New Cases)", "RR-TB %", "Frequency"),
    (plt.scatter, (v_new, v_ret), "Scatter - RR-TB % vs Estimated RR-TB Cases", "RR-TB % in New Cases",
                                  "Estimated RR-TB Cases"),
    (plt.boxplot, v_new, "Boxplot - RR-TB % (New Cases)", "", "RR-TB %")
]

# Generate plots - Histogram, Scatter, Box Plots
for func, data, title, xlab, ylab in plots:
    plt.figure()
    func(*data) if isinstance(data, tuple) else func(data, bins=20) if func==plt.hist else func(data)
    plt.title(title); plt.xlabel(xlab); plt.ylabel(ylab)
    plt.grid(True); plt.show()
```





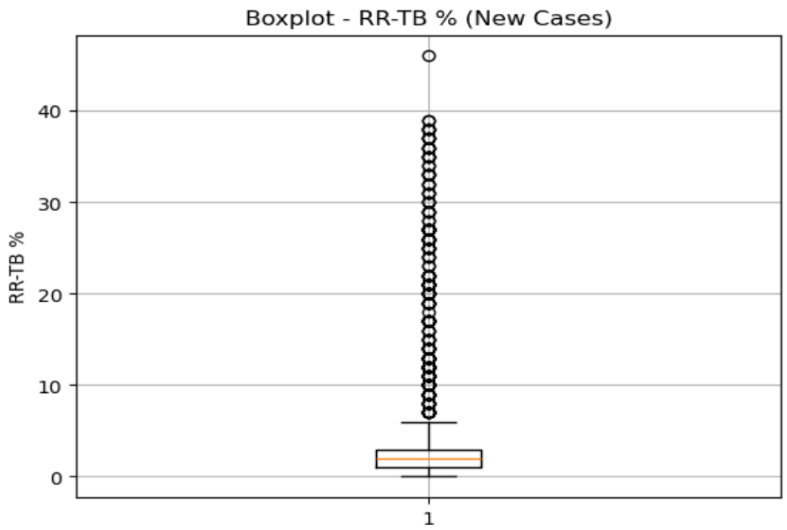


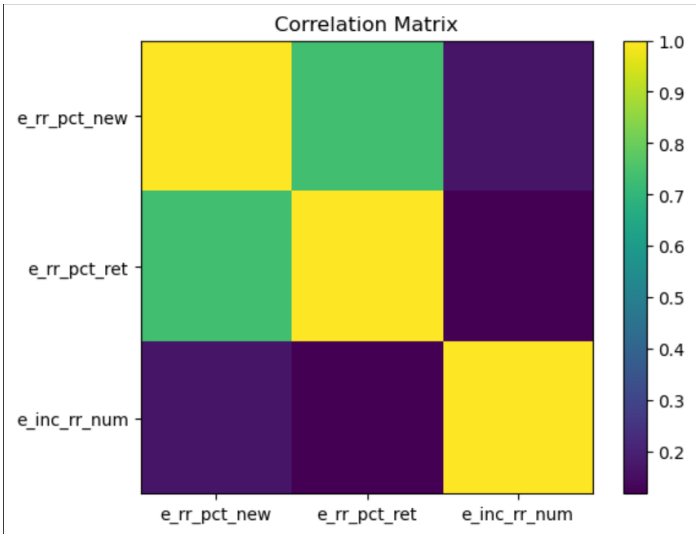
Figure 2.3.2: Visualisation of RR-TB Indicators Using Histogram, Scatter Plot, and Boxplot

Data visualisation plays a key role in presenting data analysis why because it helps to visualize patterns, distributions, and relationships within a dataset. **Figure 2.3.2** illustrates three graphical representations of the RR-TB indicators using a histogram, scatter plot, and boxplot. The histogram shows the distribution of RR-TB percentages among new cases, allowing the frequency and spread of values to be observed.

The scatter plot demonstrates the relationship between the RR-TB % in new cases and the estimated number of RR-TB cases, which helps identify possible correlations or trends between these variables. The boxplot summarises the distribution of RR-TB % by presenting the median, quartiles, and potential outliers, offering insight into the variability and spread of the data.

```
# 1. Correlation matrix - take three column names
corr = df[["e_rr_pct_new","e_rr_pct_ret","e_inc_rr_num"]].corr()
print("Correlation Matrix:\n", corr)

# Correlation Visualisation plot
plt.imshow(corr); plt.colorbar()
plt.xticks(range(3), corr.columns); plt.yticks(range(3), corr.columns)
plt.title("Correlation Matrix"); plt.show()

# 2. Outlier detection using IQR
Q1, Q3 = df["e_rr_pct_new"].quantile([0.25, 0.75])
IQR = Q3 - Q1
outliers = df.query("e_rr_pct_new < @Q1-1.5*@IQR or e_rr_pct_new > @Q3+1.5*@IQR")

print("Number of Outliers:", outliers.shape[0])
```

```
Correlation Matrix:
               e_rr_pct_new  e_rr_pct_ret  e_inc_rr_num
e_rr_pct_new       1.000000      0.732622      0.166579
e_rr_pct_ret       0.732622      1.000000      0.117669
e_inc_rr_num       0.166579      0.117669      1.000000
```



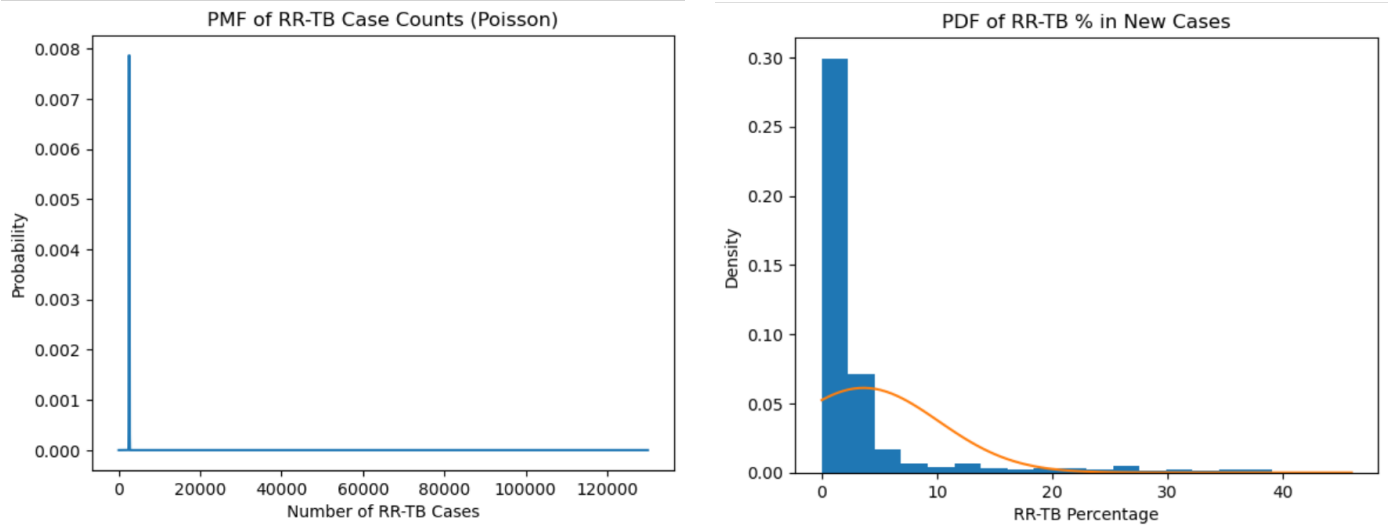

```
Number of Outliers: 193
```

Figure 2.3.3: Correlation Matrix Visualisation and Outlier Detection of RR-TB Variables

Correlation analysis was performed to explore the relationships between RR-TB variables in the dataset. A correlation matrix was calculated using the RR-TB % among new and previously treated cases and also estimated RR-TB incidence. As illustrated in above **Figure 2.3.3**, a correlation heatmap to visualise the strength and direction of these relationships. In addition, find potential outliers.

2.4 Random Variables and Probability

```python
# STEP-4 RANDOM VARIABLES & PROBABILITY
# ====================================

# Define Discrete(count data) and Continuous(percentage) Variables
X, Y = df["e_inc_rr_num"], df["e_rr_pct_new"]

# Poisson PMF (discrete) Distribution
x = np.arange(X.max()); lam = X.mean()
plt.plot(x, poisson.pmf(x, lam))
plt.title("PMF of RR-TB Case Counts (Poisson)")
plt.xlabel("Number of RR-TB Cases"); plt.ylabel("Probability"); plt.show()

# Normal PDF (continuous) Distribution
mu, sigma = norm.fit(Y); x = np.linspace(Y.min(), Y.max(), 100)
plt.hist(Y, bins=20, density=True); plt.plot(x, norm.pdf(x, mu, sigma))
plt.title("PDF of RR-TB % in New Cases")
plt.xlabel("RR-TB Percentage"); plt.ylabel("Density"); plt.show()

# Parameter summary table
results = pd.DataFrame({"Parameter":["Poisson λ","Normal μ","Normal σ"],
                        "Value":[lam, mu, sigma]})
print("DISTRIBUTION PARAMETERS"); display(results)
```



DISTRIBUTION PARAMETERS

	Parameter	Value
0	Poisson λ	2571.768802
1	Normal μ	3.635655
2	Normal σ	6.514339

Figure 2.4.1: Probability Distribution Modelling of RR-TB Data Using Poisson PMF and Normal PDF

This section examines the role of probability distributions in modelling uncertainty and showing how data values are distributed. Two types of random variables are analysed: a discrete variable representing the number of RR-TB cases and a continuous variable representing the percentage of RR-TB among new cases.

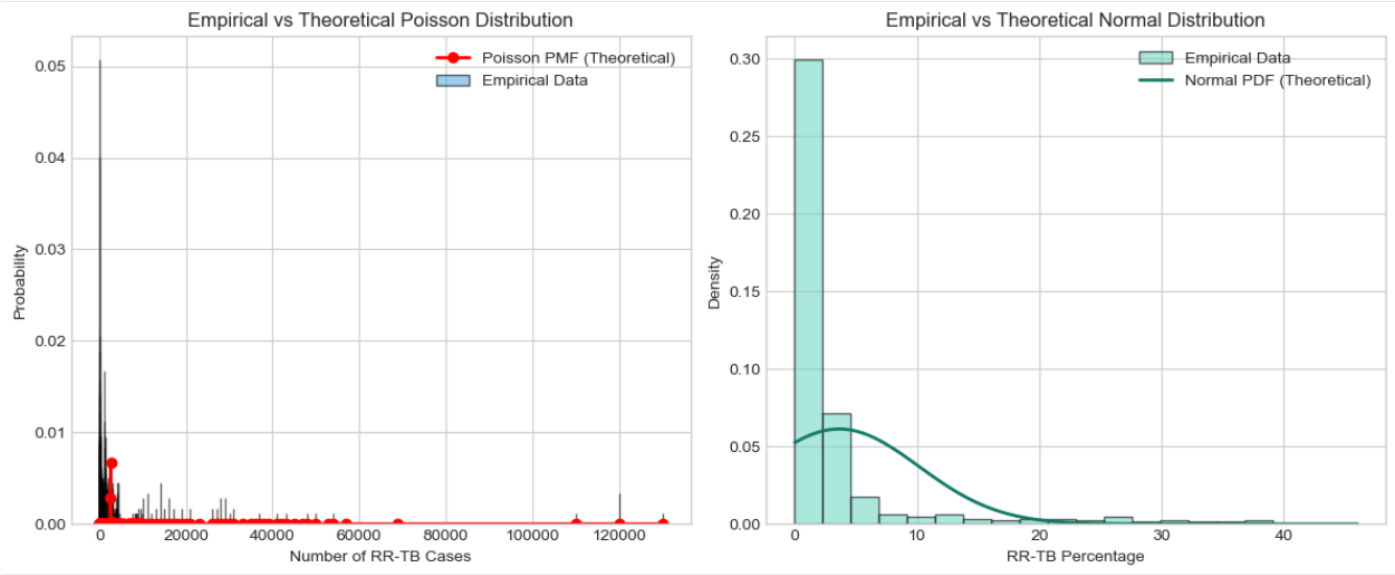
As illustrated in **Figure 2.4.1**, the Poisson distribution (PMF) was applied to the discrete count data of RR-TB cases, where the parameter λ represents the average number of cases. In also, a Normal distribution (PDF) was fitted to the continuous variable representing RR-TB %. The histogram, combined with the fitted normal curve, helps demonstrate how closely the observed data aligns with the theoretical distribution.

1. Discrete count data – RR-TB cases
2. Continuous data – RR-TB %

2.5 Probability Functions – Discrete and Continuous

```python
# STEP-5 PROBABILITY FUNCTIONS - DISCRETE AND CONTINUOUS
# ======================================================

# Select two variables

# Discrete variable (RR-TB case counts)
X = df["e_inc_rr_num"].dropna()
# Continuous variable (RR-TB percentage)
Y = df["e_rr_pct_new"].dropna()

plt.style.use('seaborn-v0_8-whitegrid')
fig, axes = plt.subplots(1, 2, figsize=(12,5))


# 1. Discrete Distribution Fit - (Poisson)
#-----------------------------------

lambda_val = np.mean(X)
values, counts = np.unique(X, return_counts=True)
empirical_prob = counts / len(X)
poisson_prob = poisson.pmf(values, lambda_val)

axes[0].bar(values, empirical_prob, alpha=0.6, color="#5DADE2",
            edgecolor="black", label="Empirical Data")
axes[0].plot(values, poisson_prob, 'ro-', linewidth=2,
             label="Poisson PMF (Theoretical)")
axes[0].set_title("Empirical vs Theoretical Poisson Distribution")
axes[0].set_xlabel("Number of RR-TB Cases")
axes[0].set_ylabel("Probability")
axes[0].legend()
```

```python
# 2. Continuous Distribution Fit - (Normal)
#----------------------------------------

mu, sigma = norm.fit(Y)
x = np.linspace(min(Y), max(Y), 100)
pdf = norm.pdf(x, mu, sigma)

axes[1].hist(Y, bins=20, density=True, alpha=0.6,
             color="#76D7C4", edgecolor="black",
             label="Empirical Data")
axes[1].plot(x, pdf, color="#117A65", linewidth=2,
             label="Normal PDF (Theoretical)")
axes[1].set_title("Empirical vs Theoretical Normal Distribution")
axes[1].set_xlabel("RR-TB Percentage")
axes[1].set_ylabel("Density")
axes[1].legend()
plt.tight_layout()
plt.show()

# Parameter summary table
results = pd.DataFrame({"Parameter":["Estimated Poisson λ","Estimated Mean","Estimated Std Dev"],
                        "Value":[lam, mu, sigma]})
print("DISTRIBUTION PARAMETERS"); display(results)
```



DISTRIBUTION PARAMETERS

	Parameter	Value
0	Estimated Poisson λ	2571.768802
1	Estimated Mean	3.635655
2	Estimated Std Dev	6.514339

Figure 2.5.1: Comparison of Empirical and Theoretical Probability Distributions for RR-TB Data

Probability distributions are used to describe the behaviour of random variables within a dataset. In this analysis, two variables were considered: the number of RR-TB cases, which was treated as a discrete random variable, and the % of RR-TB among new cases, which was treated as a continuous random variable.

As shown in **Figure 2.5.1**, the graph on the left side compares the empirical probability distribution of RR-TB case counts with a theoretical Poisson distribution, where the parameter λ represents the mean number of cases. This comparison helps determine how effectively the Poisson model represents the observed count data.

The graph on the right side illustrates the empirical distribution of RR-TB percentages using a histogram together with a fitted Normal probability density function (PDF). The parameters μ (mean) and σ (standard deviation) were estimated from the dataset to describe the central tendency and variability of the distribution.

Overall, these visualisations enable a comparison between the observed data and theoretical probability models, helping to investigates whether the chosen distributions accurately reflect the characteristics of the dataset or not.

2.6 Statistical Distribution Check

```
# Empirical Validation of Theoretical Distribution Assumptions
#-------------------------------------------------------------
a = 0.05

exp = poisson.pmf(values, lambda_val)*len(X)
obs, exp = counts[exp>=5], exp[exp>=5]
exp *= obs.sum()/exp.sum()
chi,p = chisquare(obs,exp)

# GOF - Goodness-of-Fit

# Poisson test
print("Poisson GOF (α=0.05)\nChi²=%.3f p=%.5f"%(chi,p))
print("Evidence against H0\n" if p<a else "No evidence against H0\n")

# Normality test
ks,p = kstest(Y,'norm',args=(mu,sigma))
print("Normality Test (α=0.05)\nKS=%.3f p=%.5f"%(ks,p))
print("Evidence against H0" if p<a else "No evidence against H0")
```

```
Poisson GOF (α=0.05)
Chi²=5.624 p=0.01772
Evidence against H0

Normality Test (α=0.05)
KS=0.331 p=0.00000
Evidence against H0
```

Figure 2.6.1: Goodness-of-Fit Testing for Poisson and Normal Distribution Models

Finally, whether the selected probability distributions accurately represent the dataset, Goodness-of-Fit (GOF) tests were conducted. The Chi-Square test was used to evaluate the suitability of the Poisson distribution for modelling the discrete variable that represents RR-TB case counts. This above test compares the observed frequencies in the dataset with the expected observations calculated from the theoretical Poisson distribution.

In addition, the Kolmogorov–Smirnov (KS) test was applied to assess whether the continuous variable representing RR-TB percentages follows a normal distribution. This test evaluates the difference between the empirical distribution vs theoretical normal distribution.

Both tests were carried out using a significance level of $\alpha = 0.05$. The resulting p-values indicate whether there is enough statistical evidence to reject the null hypothesis that the data follows the assumed distribution. These inferential tests therefore provide statistical support for evaluating the appropriateness of the probability models used in the analysis.

Task 3: Critical Reflection

3.1 Mathematical Representation of the Dataset

Mathematical techniques such as vectors, matrices, descriptive statistics, and visualisations aligned well with the dataset. These methods helped organise the data structure and identify relationships between RR-TB indicators across countries. The goodness-of-fit tests also suggested that the Poisson and Normal distributions reasonably represent the dataset, supporting the use of probability models in the analysis.

3.2 Limitations of Statistical Assumptions in Real-World Data

Although mathematical models were useful, some theoretical assumptions may not fully reflect real-world conditions. Assumptions such as independence, normality, and randomness can be difficult to satisfy in global health datasets because TB data may be influenced by socioeconomic conditions, healthcare access, and reporting differences between countries. These factors can affect the accuracy of statistical interpretations.

3.3 Potential Misinterpretation of Mathematical Techniques

Improper use of mathematical tools can lead to misleading conclusions in data analysis. If data scientists rely on statistical models without checking assumptions or validating results, the analysis may produce inaccurate insights. Therefore, careful interpretation and validation of mathematical models are essential to ensure reliable and meaningful conclusions when analysing complex datasets.

References

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